

AYMANE AIT DADS

AI Software Intern | Agentic AI · LLM Fine-Tuning · Intelligent Automation

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by architecting an end-to-end LoRA SFT pipeline on a 30B-parameter Mamba-Transformer with BF16 precision and 8,192-token context. Reduced manual network analysis time by 60% at Orange Maroc by deploying automated ML pipelines on 100K+ records, translating model outputs directly into stakeholder-ready maintenance recommendations. Brings hands-on expertise in agentic pipelines, LLM orchestration, RAG, tool-calling, and LangChain - skills directly aligned with UL Solutions' focus on agent-based AI solutions for complex enterprise workflows. Aims to prototype and evaluate intelligent automation systems at UL Solutions that advance reasoning-capable, production-ready agentic AI across enterprise software platforms.

CORE COMPETENCIES

Agentic AI Systems | Large Language Models | Intelligent Automation | LLM Orchestration | Prototype & Evaluate | Reasoning & Orchestration | Enterprise Software Platforms | Agent-Based AI Solutions

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Automated feature engineering and model evaluation workflows using Python and Scikit-learn, reducing manual analysis time by ~60% across 100K+ fiber-optic network records.
- Built a **Random Forest + K-Means clustering pipeline** that classified network quality across thousands of nodes and mapped customer performance profiles to 5 commercial service tiers.
- Delivered stakeholder presentations translating model outputs into **actionable maintenance recommendations** adopted directly by the network optimization team.
- Developed end-to-end data pipeline covering ingestion, feature engineering, and Power BI visualization for **upload speed, latency, and jitter** metrics at production scale.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge - Kaggle (In Progress)

Kaggle Competition · 2025-2026 · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context, targeting all attention + MLP + Im_head projections) using Unsloth and SFTTrainer.
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers producing 605 and 64 verified chain-of-thought rows respectively.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton

AI-Generated Image Detection - NTIRE 2026 @ CVPR

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in EURECOM class** (private Kaggle leaderboard, 0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M parameters) with a custom MLP head, dual learning rates, and FP16 mixed precision on 250K images spanning 25+ generator types.
- Submitted independently to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; discovered that 1-epoch training (0.793 AUC) outperforms 4-epoch (0.767 AUC), demonstrating stronger out-of-distribution generalization via early stopping and 10-view TTA ensemble.

PyTorch · OpenCLIP · Hugging Face Transformers · AdamW · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-Level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Cloud Computing, Statistics, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

LLM & Agentic AI: LoRA / QLoRA SFT | LLM Orchestration | Agentic Pipelines | RAG | Tool-Calling | LangChain | Prompt Engineering
ML Frameworks & Engineering: PyTorch | Hugging Face Transformers | Unsloth | TRL | PEFT | vLLM | Scikit-learn
Data, MLOps & Cloud: Python | Pandas | SQL | Git | Docker | Supabase | REST APIs | Power BI